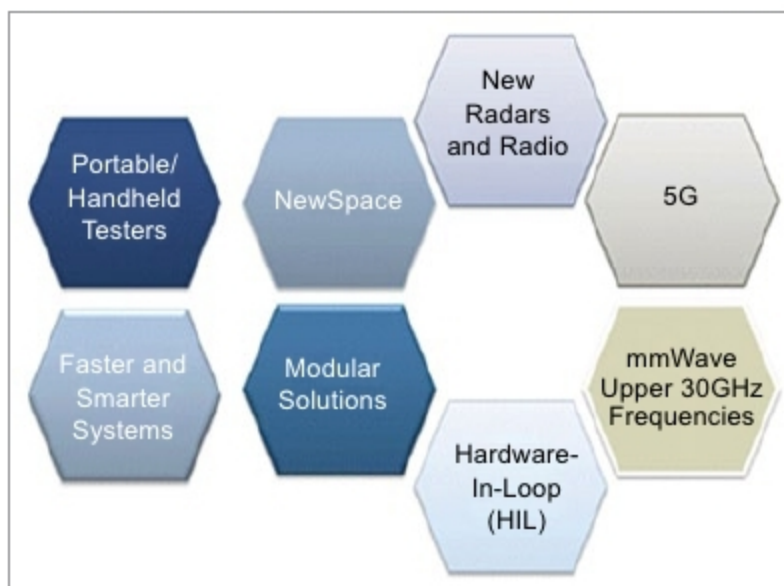




Technology trends impacting the electronic test equipment market in defence (Credit: Frost & Sullivan)

radar and other electronic warfare technologies that have more signal processing capabilities. These technologies require T&M equipment capable of processing large RF bandwidths. Demand for products that can address high bandwidth and signal processing requirements for high-bandwidth data is expected to grow.

Sandeep Kapoor, regional marketing head - EMEA and India, at Keysight Technologies, says, "Growing demand for multi-mission capable military and defence machinery and system electronics requires us to deploy complex systems that have a broader bandwidth, wide frequency range and real-time processing capacity. Radar systems, communications on-the-move and satellite innovations are driving the need for higher frequencies and more bandwidth to support uninterrupted connections. This requires precise measurement, analysis and interpretation of results in new spectrum environments.

"Many technologies in radar, electronic warfare and signal intelligence are moving to multiple distributed apertures to deliver higher performance and support for emerging beamforming technologies. This requires synchronisation of multiple RF/microwave channels for communication and sensing. As commercial technologies are adapted for cyber-attacks, every decibel of sensitivity counts to improve the stand-off range and countermeasures. There is a need to improve sensitivity to be able to better monitor airwaves,

and intercept and deflect increasingly-difficult-to-find signals."

With the arrival of 5G, pushing the requirements of signal and spectrum analysers, new multichannel applications will have a direct impact on bandwidth usage and frequency ranges. Drastic shifts in signal generators, signal and spectrum analysers, network analysers and power meters of the electronics T&M market in R&D as well as manu-

facturing applications are expected to occur by 2020.

Software-defined tools

Software-defined radios (SDRs) and software-defined instrumentation (SDI) are ramping up software-defined testing to add flexibility in markets where products and protocols are evolving or still being defined, and where system architectures are being tweaked or replaced to deal with an explosion of data. In effect, the entire compute infrastructure across multiple markets is shifting, and the number of signals that need to be optimised and processed is rising significantly.

Software-based instrumentation builds similar levels of flexibility as conventional T&M by not relying on benchtop, handheld and standalone instruments. Software-defined testing also provides the ability to do more tests using a single testing device.

SDI and SDR are used in a wide range of applications, from general-purpose T&M applications to advanced wireless research. In more traditional T&M applications, technical benefits of a software-centric approach and the fast pace of change in the wireless industry are driving widespread adoption. More specifically, the fast evolution of new wireless technologies like 802.11ax and 5G are driving engineers to use instrumentation that is more flexible, especially because the cost of updating the hardware's software is significantly less than that of purchasing new hardware.

SDI and software-defined testing are now a requirement across an increasing number of markets. That trend will only grow across new markets as the quantity of data increases and the need for moving that data increases with it.

Challenges and improvement areas

T&M equipment suppliers are required to deliver high-performance measurement solutions to cope with the increase in complexity in military electronic systems. Greater complexity in military electronic systems requires more capable T&M equipment to test those systems. Increased use of robotics, automated systems, and smaller and lightweight radios requires updated multiple-frequency bands, modulation formats and digital electronic systems.

Emerging test challenges are stimulating over-the-air sensors that eliminate physical contact to the device under test. Real-time streaming is another military concept that needs to be catered to. Increasing use of military unmanned aerial vehicles and drones is the main driver behind the expected growth in demand for electronic test equipment aimed at military systems.

Future scope

Increasing device complexity and time-to-market pressure will eventually lead to the growth of SDI systems. SDR measurements are new developments in the field of communication technologies. SDR designs require greater integration of digital signal processors.

Network-centric warfare systems are expected to be integrated, thereby connecting more platforms across the network to support high-volume data and cleared volume transmission. Need for high-throughput connectivity is expected to create a demand for upgrading systems that are compatible with increased data rate needs.

An increase in R&D investments as well as government spending in upgrading the systems and their capabilities will create significant revenue opportunities for T&M vendors. **EFY**